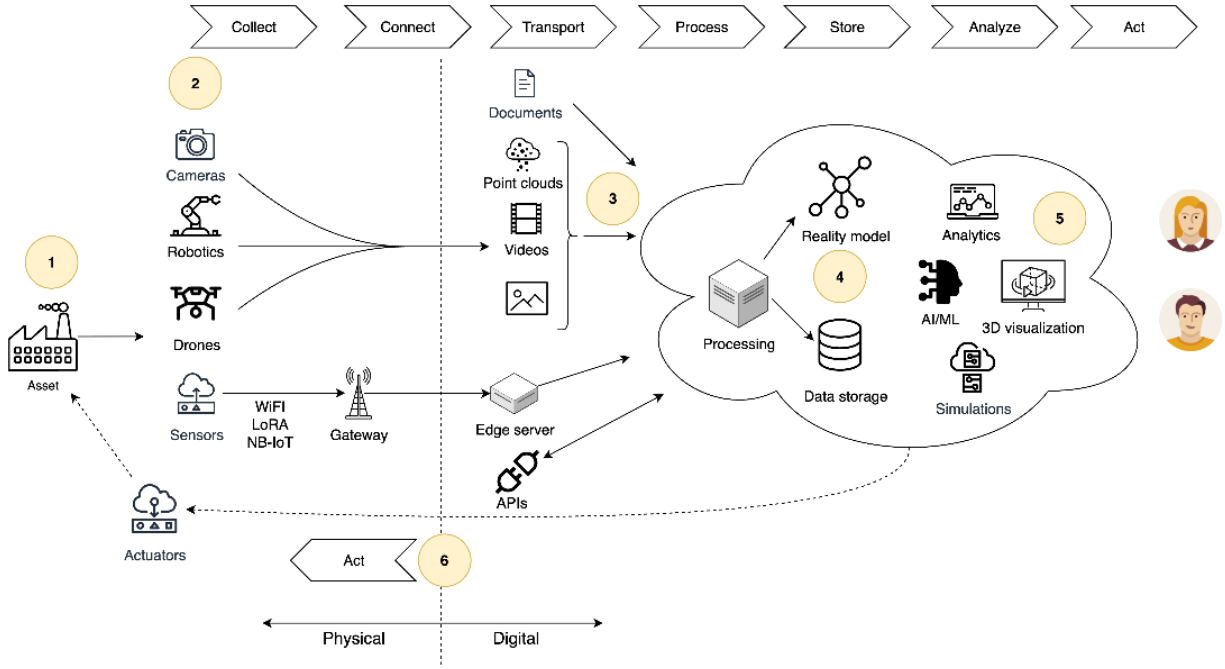


Figure 1.12 High level architecture of a digital twin showing how data about the real world is collected, stored, and processed to make decisions and affect outcomes.

This figure illustrates the complete digital twin data flow, showing how information moves from physical assets through various collection and processing stages to enable intelligent decision-making and automated actions. The numbered elements trace the journey from initial data capture to final control actions.

1. Understand the object or system that you plan to build the twin of, and critically, what specific objectives you hope to achieve.
2. Determine what data you will need to collect to create a digital model that will support your desired objectives. Begin with static data sources, including documents and still images, before connecting sensors, cameras, and external systems via APIs.
3. Connect these data sources to your digital environment and transport the data into it.
4. Process data sources to build a model of the physical system within the digital environment. This model may include visual models, mathematical models that define your system, and a model of entities and their relationships persisted in a range of data stores.
5. Run analytics and simulations over the model.